README: NS-3 Task 2 Simulation

1. Project Overview

This project simulates a TCP vs. UDP bottleneck comparison using star topology in NS-3. It demonstrates TCP starvation when competing with non-responsive UDP traffic over a limited bandwidth link.

2. Files Included

- **scratch/task2.cc**: The main C++ simulation source code.
- **plot_results.py**: A Python script to parse trace files and generate graphs.
- **task2.xml**: The NetAnim animation file (generated after running the simulation).
- ***.dat files**: Raw trace data files (generated after running the simulation).

3. Prerequisites

- **NS-3 Simulator** (v3.30 or newer recommended).
- **Python 3** with matplotlib installed for plotting.
  - *To install matplotlib:* pip install matplotlib
- **NetAnim** (Optional, for visual replay).

4. How to Run the Simulation

Step 1: Place the source file

Copy the task2.cc file into the scratch directory of your NS-3 installation.

Bash

```
cp task2.cc ns-3-dev/scratch/
```

Step 2: Build and Run

Open a terminal in your main ns-3-dev directory and run the simulation using the Waf/ns3 wrapper:

Bash

```
./ns3 run scratch/task2
```

Expected Output:

The console will display the end-to-end flow statistics (Tx/Rx packets, Throughput, and Packet Loss).

Example:

Flow UDP (n0->n3): TxPackets: 439 ... LostPackets: 0

Flow TCP (n1->n3): TxPackets: 807 ... LostPackets: 54

Generated Files:

After the simulation finishes, the following files will appear in your root directory:

- udp_throughput.dat
- tcp_throughput.dat
- queue_occupancy.dat
- drops.dat
- task2.xml

5. How to Reproduce the Plots

Step 1: Run the Python Script

Ensure the plot_results.py file is in the same directory as the generated .dat files. Run the script:

```
Bash
python3 plot_results.py
```

Step 2: View Graphs

The script will generate two image files in the current directory:

1. **throughput_graph.png**: Shows TCP (Blue) vs. UDP (Red) throughput over time.
2. **queue_graph.png**: Shows the queue size variations at the bottleneck router ($n2$).

6. Visual Replay (NetAnim)

To view the simulation topology and packet flow visually:

1. Launch NetAnim:

```bash
./NetAnim
```

(Note: Depending on your install, this might be inside netanim/ folder, e.g., ../netanim/NetAnim)

2. Click the **Open** folder icon.
3. Select the **task2.xml** file generated in the previous steps.
4. Click the **Play** button.
    a. *Note: The nodes are labeled: n0 (UDP Source), n1 (TCP Source), n2 (Router), n3 (Sink).*